

Compare Ratios

Lesson 3-5

Name: _____

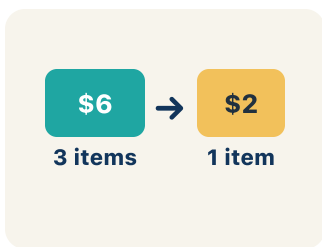
Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



\$3 per 1 pound → unit rate is \$3/lb

Unit rate

Write the definition:

$$2:3 = 4:6$$

$\times 2 \rightarrow$ same comparison

4:6 and 2:3 are equivalent (both simplify to 2:3)

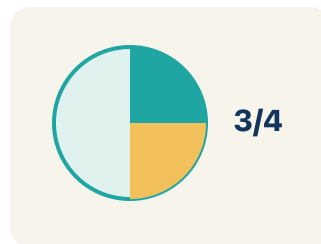
Equivalent ratio

Write the definition:

3:4 vs. 2:5 — which has more of the first ingredient per unit?

Compare

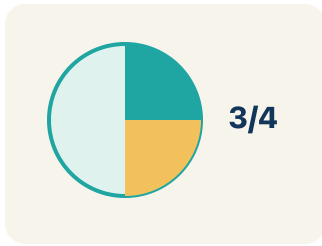
Write the definition:



8:12 → divide both by 4 → 2:3

Simplify

Write the definition:



To compare $\frac{3}{5}$ and $\frac{4}{7}$, use LCD 35: $\frac{21}{35}$ vs $\frac{20}{35}$

Common denominator

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can compare ratios by finding unit rates or using equivalent ratios.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Unit rate

Equivalent ratio

Compare

Simplify

Common denominator

1 A rate that tells the amount for exactly one unit, like 5 miles per 1 hour, is a _____.

2 _____ — Two ratios that mean the same thing.

3 To decide which ratio is greater or if they are equal is to _____ them.

4 Writing a ratio in its smallest equal form is to _____ it.

5 A denominator shared by two fractions so they can be compared is a _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: Chef A uses 2 cups of cheese for every 8 crackers. Chef B uses 3 cups of cheese for every 9 crackers. Who uses more cheese per cracker?

- A. Chef B
- B. Chef A
- C. They use the same amount
- D. Cannot determine

Step 1 Chef A: $2 \div 8 = 0.25$ cups per cracker.

Step 2 Chef B: $3 \div 9 \approx 0.33$ cups per cracker.

Step 3 Chef B uses more cheese per cracker.

 **Answer:** A. Chef B

 We do — together

Solve this one with your class using the same steps.

Problem: Which ratio is greater: 3:4 or 5:8?

- A. 3:4
- B. 5:8
- C. They are equal
- D. Cannot determine

Step 1 _____

Step 2 _____

Step 3 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Which lemonade recipe is more lemony? Recipe A: 2 lemons for 6 cups of water. Recipe B: 3 lemons for 7 cups of water.

- A. Recipe B
- B. Recipe A
- C. They are the same
- D. Cannot determine

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Two soup recipes use salt. Recipe A uses 4 tsp salt for 10 cups of broth. Recipe B uses 3 tsp salt for 8 cups of broth. Which soup is saltier per cup?

- A. Recipe B
- B. Recipe A
- C. They are equally salty
- D. Cannot determine

Show your work:

2. Chef Mara's lemon glaze uses 6 oz of lemon juice for 9 cupcakes. Chef Devi's uses 4 oz for 7 cupcakes. Whose glaze has more lemon per cupcake?

- A. Chef Mara
- B. Chef Devi
- C. Same amount
- D. Cannot determine

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Store A sells 5 notebooks for \$12. Store B sells 8 notebooks for \$20. Without calculating, a student says Store B must be cheaper because you get more notebooks. Is the student's reasoning valid? Find the unit rates to prove your answer.

Show your work:

Reflect — Exit Ticket

A bakery sells 4 muffins for \$10 and another bakery sells 6 muffins for \$14. Which bakery has the lower price per muffin?

- A. The second bakery
- B. The first bakery
- C. Same price
- D. Cannot determine

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Unit rate
2. **Notes 2:** Equivalent ratio
3. **Notes 3:** Compare
4. **Notes 4:** Simplify
5. **Notes 5:** Common denominator
6. **We Try (notes):** A. $3:4$ — *Convert to the same denominator: $3:4 = 6:8$. Since $6:8 > 5:8$, the ratio $3:4$ is greater.*
7. **You Try (notes):** A. Recipe B — *Recipe A: $2 \div 6 \approx 0.33$ lemons per cup. Recipe B: $3 \div 7 \approx 0.43$ lemons per cup. Since $0.43 > 0.33$, Recipe B is more lemony.*
8. **Try It 1:** A. Recipe B — *Recipe A: $4 / 10 = 0.4$ tsp per cup. Recipe B: $3 / 8 = 0.375$ tsp per cup. Since $0.4 > 0.375$, Recipe A is saltier per cup.*
9. **Try It 2:** A. Chef Mara — *Chef Mara: $6 / 9 = 0.666\dots$ oz per cupcake. Chef Devi: $4 / 7 = 0.571\dots$ oz per cupcake. Since $0.67 > 0.57$, Chef Mara's glaze has more lemon per cupcake.*
10. **Exit Ticket:** A. The second bakery — *First bakery: $\$10 \div 4 = \2.50 per muffin. Second bakery: $\$14 \div 6 \approx \2.33 per muffin. The second bakery has the lower price.*